User inputs and decision points

Confirm suggested defaults or specify alternatives.

1 Computing environment

Which computing cluster and calculation root?
Specify the cluster and absolute working directory.

Which Python virtual environment?
Use an existing environment, or agree to create one.

Which VASP executable and launch command?
Use a compatible installed build.

Which submission script and resources?
Provide the scheduler script and resource settings.

Where are the pseudopotential datasets?
Configure the path to the licensed library.

2 Research scope

Which compound and crystal phase?
Specify the formula and an identifier when available.

Where should the unit-cell structure come from?
Provide a CIF or POSCAR, or use a materials database.

Intrinsic defects, dopants, or both?
Generate intrinsic defects when requested.

Which dopant species, if any?
Required only when dopants are included.

3 Computational details

What is the plane-wave cutoff energy?
Suggested default: 400 eV; specify another if preferred.

What is the electronic dielectric method?
Suggested default: GGA-PBE; hybrid functional optional.

What supercell size should be used?
Suggested default: over 200 atoms; supercell matrix optional.

How should defect charge states be selected?
Suggested default: doped recommendations; or supply an integer list or inclusive range.

4 Thermodynamic analysis

What synthesis temperature?
Suggested default: 1000 K.

What quench target temperature?
Suggested default: 300 K.

For dopants, which reservoir conditions?
Needed only when dopant chemical potentials are required.

Accept doped recommendations collectively; specify custom charge states one defect species at a time.